Exercises Week 5

Raul Pekander

October 8, 2026

Setup

Let's use the following discretisation t = ih, h = 0.01, i = 0,...,100 in dimension one

$$t_i \in [0, 1] \tag{1}$$

The distance D_{ij}

$$D_{ij} = ||t_i - t_j||_2 \tag{2}$$

The Matérn covariance

$$\begin{aligned} C_v(d_{ij}) &= \sigma^2 \frac{2^{1-v}}{\Gamma(v)} \left(\frac{\sqrt{2v}d}{l} \right)^v K_v \left(\frac{\sqrt{2v}d}{l} \right), i \neq j \\ C_v(d_{ii}) &= \sigma_f^2, \forall i \in \{1, ..., n\} \end{aligned} \tag{3}$$

(Rasmussen, 2005)
Let K_θ represent a discretised boxcar kernel with a width θ of an unit area

$$K_{\theta,ij} = \frac{h}{\theta} \text{ if } D_{ij} \leq \theta/2 \text{ otherwise } 0 \tag{4}$$

For the simulation we use Cholesky decomposition $C = LL^T$ and we draw a realization $x_{true} = Lz, z \sim N(0, I)$ so that $y = K_{\theta_{true}} x_{true} + e$ (Pourahmadi, 2014).

Task 1

The Matérn parameters $[\nu, l, \sigma_n, \sigma_f] = [3/2, 0.40, 0.10, 1.0]$.
The posterior is $p(x, \theta|y) \propto p(y|x, \theta)p(x)p(\theta)$. Dropping the constant and taking negative logarithm gives

$$J(x, \theta) = \frac{1}{2\sigma^2}||y - K_\theta x||^2 + \frac{1}{2}x^T C^{-1} x - \log p(\theta) \tag{5}$$

$\theta \sim U(a, b)$. Let the unknown be a vector (x,θ). X is unbounded and θ is bounded be a and b. Then the gradient on is

$$\nabla_x J = -\frac{1}{\sigma^2} K_\theta^T (y - K_\theta x) + C^{-1} x \tag{6}$$

At θ we cannot differentiate the boxcar so instead we calculate dJ/dθ with finite difference.
Then L-BFGS-B is run with initial value $x_0 = 0$ and $\theta_0 \in (a, b)$. We obtain the estimates $(\hat{x}, \hat{\theta})$.(Andrei, 2022).

Conditional of x Given a fixed θ

$$p(x|\theta, y) \propto \exp\left(\frac{1}{2\sigma^2}||y - K_\theta x||^2 - \frac{1}{2}x^T C^{-1} x\right) \tag{7}$$

It follows that

$$\begin{aligned} x|\theta, y &\sim N(m_\theta, A_\theta^{-1}) \\ A_\theta &= \frac{1}{\sigma^2}K_\theta^T K_\theta + C^{-1} \\ m_\theta &= \frac{1}{\sigma^2}A_\theta^{-1}K_\theta^T y \end{aligned} \tag{8}$$

Conditional of θ We sample with Metropolis-Hasting

$$p(\theta|x, y) \propto p(\theta)\exp\left(-\frac{1}{2\sigma^2}||y - K_\theta x||^2\right) \tag{9}$$

We sample x using Gibbs sampler and θ with Metropolis-Hastings.
For x:

$$\begin{aligned} &\text{Let } \theta^{(0)} = \hat{\theta}. \\ &\text{We will iterate from i to N.} \\ &\text{1.step: } A = A_\theta, m = m_\theta \text{ as } \theta = \theta^{(k-1)} \\ &\text{2.step: } A = RR^T \\ &\text{3.step: } x^{(k)} = m + R^{-T}z, z \sim N(0, I) \end{aligned} \tag{10}$$

For θ:

$$\begin{aligned} &\text{Let } \theta^{(0)} = \hat{\theta}. \\ &\text{We will iterate from i to N.} \\ &\text{1.step: } \theta' = \theta^{(k-1)} + s\xi, \xi \sim N(0, 1) \\ &\text{2.step: } \log\pi(\theta) = -\frac{1}{2\sigma^2}||y - K_\theta x^{(k)}||^2 + \log p(\theta) \\ &\text{3.step: accept the proposal } \theta' \text{ with probability} \\ &\alpha = \min\left\{1, \exp(\log(\pi(\theta') - \log\pi(\theta^{(k-1)})\right\} \end{aligned} \tag{11}$$

Exercise 2

In order to have a smooth kernel, we can use a Gaussian process prior, $k \sim N(0, C_k)$.
The negative log-posterior and its gradient are:

$$\begin{aligned} J(k) &= \frac{1}{2\sigma^2}\sum_{i=1}^{M} ||y_i - X_i k||^2 + \frac{1}{2}k^T C_k^{-1} k \\ \nabla J(k) &= -\frac{1}{\sigma^2}\sum_{i=1}^{M} X_i^T (y_i - X_i k) C_k^{-1} k \end{aligned} \tag{12}$$

To sample k with Gibbs, let set $A = \frac{1}{\sigma^2}\sum_{i=1}^{M} X_i^T X_i + C_k^{-1}$ and $b = \frac{1}{\sigma^2}\sum_{i=1}^{M} X_i^T y_i$. By consequence, $p(k|data) \propto \exp(-\frac{1}{2}k^T Ak + b^T k)$ and $k|data \sim N(A^{-1}b, A^{-1})$. The result from L-BFGS should be $A^{-1}b$. (Calvetti and Somersalo, 2023)
The full conditional is:

$$k_j|k_{-j}, data \sim N\left(\frac{b_j - \sum_{l\neq j} A_{jl}k_l}{A_{jj}}, \frac{1}{A_{jj}}\right) \tag{13}$$

Metrics

Acceptance rate for theta	0.33
ESS of theta	137.226 / 8000
RMSE_x posterior mean	0.191
RMSE_x l-bfgs	0.188

Table 1: Exercise 1 metrics

ESS of theta	220.602/5000

Table 2: Exercise 2 metrics

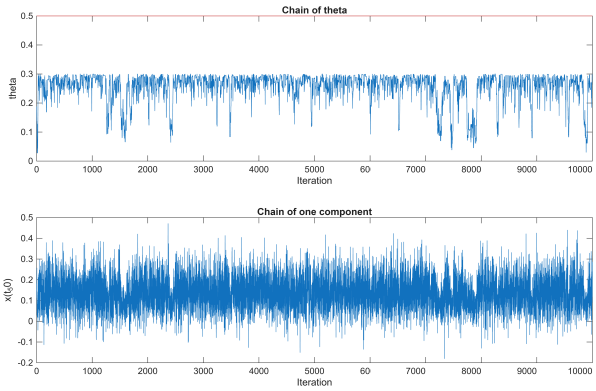
References

Andrei, N. (2022). *Modern numerical nonlinear optimization.* Springer Optimization and Its Applications ; v.195. Springer, Cham, Switzerland.

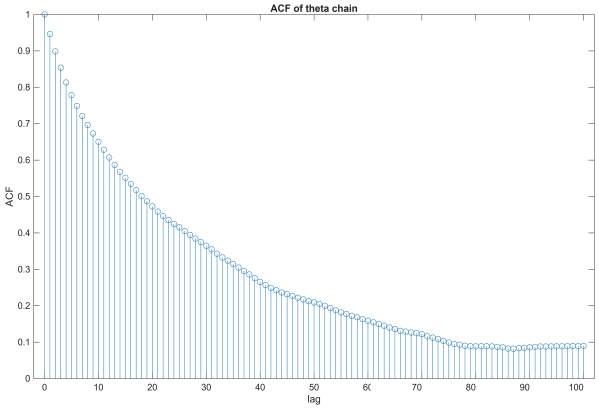
Calvetti, D. and Somersalo, E. (2023). *Bayesian Scientific Computing.* Applied Mathematical Sciences, 215. Springer International Publishing, Cham, 1st ed. 2023. edition.

Pourahmadi, M. (2014). *High-Dimensional Covariance Estimation: With High-Dimensional Data.* Newark: John Wiley et Sons Ltd.

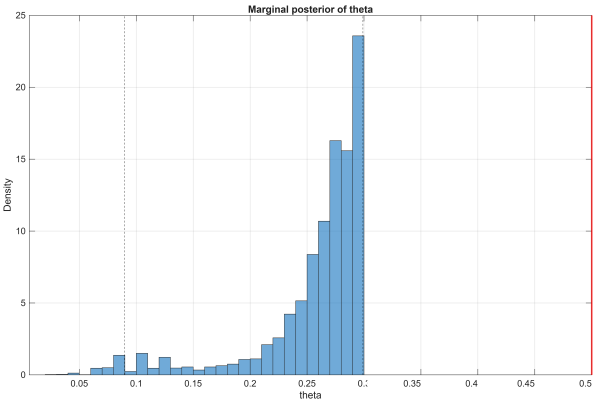
Rasmussen, Carl Edward, W. C. K. (2005). *Gaussian Processes for Machine Learning.* MIT Press.



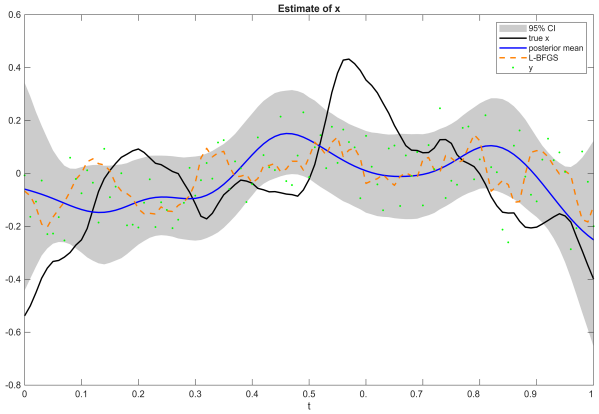
(a) Exercise 1:Chain of theta



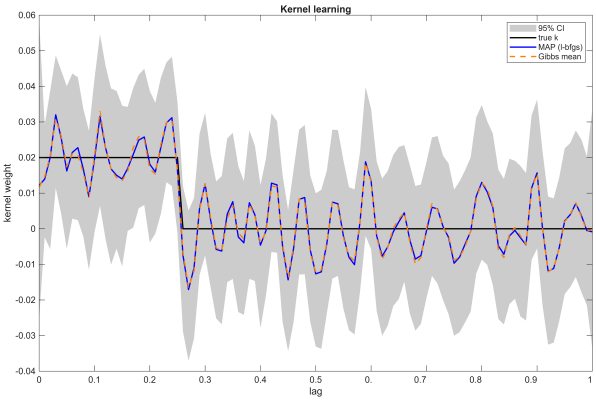
(b) Exercise 1:ACF of the chain



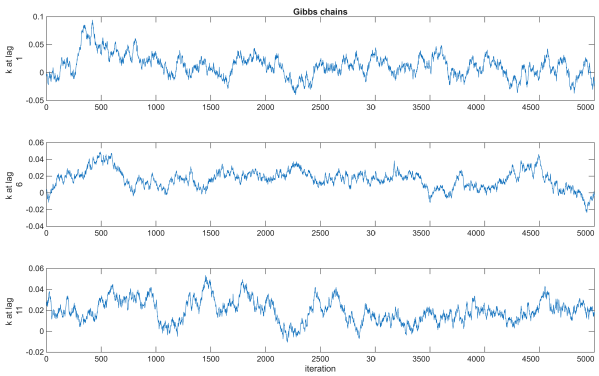
(c) Exercise 1: Marginal posterior of theta



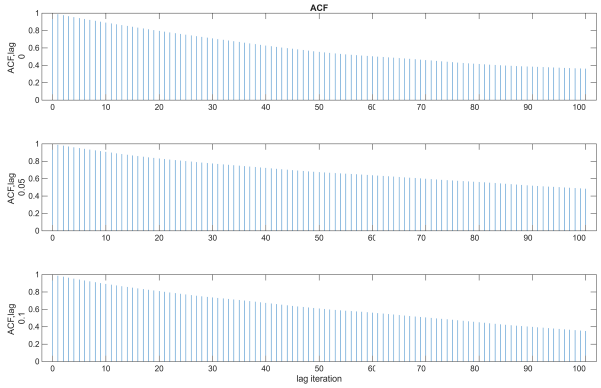
(d) Exercise 1: Estimate of x



(e) Exercise 2: estimates



(f) Exercise 2: Chains



(g) Exercise 2: ACF of the chains